

Task 1: Calculate a car's average consumption being provided the total distance traveled (in Km) and the spent fuel total (in liters).

Sample Input	Sample Output
500 35.0	14.286 km/l
Explanation: First input 500 represents the total distance (in Km) and the second one is a floating point number representing the spent fuel (in Litre).	

Task 2: Write a Java program that will ask the user for the first value, change, and last value of an arithmetic series, and then print the series.

Sample Input	Sample Output
10 20 134	10, 30, 50, 70, 90, 110, 130
Explanation: First value = 10, change = 20, last value = 134. The terms of the series increase by 20, and stop at 130 because 140 would exceed 134. You have to print the commas as well.	
Sample Input	Sample Output
0 100 1234	0, 100, 200, 300, 400, 500, 600, 700, 800, 900, 1000, 1100, 1200

Task 3: Write Java code of a program that asks the user the value n as input and then prints the value of y. **Hint:** Every 4th term is a negative number.

$$y = 1 + 1/2 + 1/3 - 1/4 + 1/5 + 1/6 + 1/7 - 1/8 + \dots 1/n$$

Sample Input	Sample Output
4	Y = 1.5833
Explanation: The result is calculated according to the formula.	

Task 4: Write a Java code that asks the user for the value of N as input and then prints the value of Y.

$$y = 3 - 5 + 7 - 9 \dots \text{nth term}$$

Sample Input	Sample Output
5	7
Explanation: If the user gives n=1, print 3. If the user gives n=2, print -2. If the user gives n=3, print 5 and so on. For example, for the input 5 we'll add up the first 5 numbers of the series, so 3-5+7-9+11 = 7.	

Task 5: Write a java program that takes an integer input from the user and calculates the sum of its digits. If the summation is even, please print "The number is even." otherwise print "The number is odd."

Sample Input	Sample Output
54637	The sum is odd.
Explanation: 5+4+6+3+7=25 which is odd.	
2754	The sum is even.
Explanation: 2+7+5+4=18 which is even.	

Task 6: Write a java program that takes a binary number as input from the user and prints its decimal equivalent.

Sample Input	Sample Output	Explanation
1111	15	$1(2^3) + 1(2^2) + 1(2^1) + 1(2^0) = 15$
0101	5	$0(2^3) + 1(2^2) + 0(2^1) + 1(2^0) = 5$

Arrays + Strings + Methods

Task 7

A. Write a method called **isVowel** which takes a string in its parameter and counts all the vowels in the String. If any vowel exists in the string then the method returns the **count**.

Sample Input	Sample Output
The quick brown fox jumps over the lazy dog	Number of vowels in the string: 11

B. Write a method called **isConsonant** which takes a string in its parameter and counts all the consonants in the String. If any consonant exists in the string then the method returns the **count**.

Sample Input	Sample Output
The quick brown fox jumps over the lazy dog	Number of consonants in the string: 24

C. Write a method called **vowel/consonantSum** which takes an array of strings in its parameter and returns the summation of the number of vowels/consonants.

Note: **You must call** the methods written in tasks A/B, otherwise this task will be **considered invalid**.

Given Array	Sample Output
<pre>String [] names = {"Bob", "Alice", "Max", "Marry", "Rosy"}; System.out.println("The total number of vowels in the array is:" + vowelSum(names)); System.out.println("The total number of consonants in the array is:" + consonantSum(names));</pre>	The total number of vowels in the array is: 7 The total number of consonants in the array is: 13

[There are more coding problems in the actual mid review lab file.]